Advising Philosophy

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This document focuses on my thoughts on supervising PhD students. Some of them also apply to supervising postdoc researchers.

1 Why I am writing

- **Clarify my thinking.** Writing helps me turn ambiguity into clarity. I want the thoughts in this document to guide my conduct and give me a standard to return to as an advisor.
- **Learn from experience.** Experiences of friction or a poor fit have often led me to reflect more deeply on what students and advisors need from each other. Keeping this document up to date helps me refine my thinking as I learn.

I am still relatively early in my advising career and continue to learn through practice. I was fortunate to have excellent mentors during my own training, and I feel a responsibility to pass on some of the guidance and support I received to the next generation.

2 What I find rewarding about advising

Advising doctoral students is one of the most important reasons I chose an academic career. Four aspects of the student-advisor relationship are especially rewarding to me.

- **Accomplishing meaningful work together.** I enjoy working side by side with students to explore the unknown and create something new. Pursuing an important yet difficult research question together and making a meaningful contribution bring a shared sense of accomplishment. This is a fun and adventurous journey.
- **Seeing students grow.** It is deeply rewarding to see students develop their abilities, confidence, and independence over time. Students often begin by learning from their advisors. By the end of the PhD, they should surpass their advisors in their areas of specialization. Helping them make that progress and seeing their growth is one of the great joys of advising.
- **Deepening my own understanding.** Pursuing wisdom, clarity, and understanding is a central goal in my life, and I find the pursuit itself deeply rewarding. Students' insights help me understand the subjects we study more clearly. Our interactions also reveal how different people think, a fascinating experience that helps me grow as an advisor and as a person.
- **Seeing ideas carried forward.** I hope that, over time, students may build on some of what I have come to understand, develop it in their own ways, and pass it on to others. The possibility that these ideas might grow beyond my own work is meaningful to me. This is a hope, not an obligation; students should be free to choose their own paths.

These rewards depend on both students and advisors taking their responsibilities seriously and following through on their commitments. Trust develops over time, and I hope these relationships can eventually grow into lasting friendships.

3 Finding a good fit

Good advising begins with a match in interests, preparation, and ways of working. An excellent student may still be a poor fit for a particular advisor. Students should also consider whether my research interests, expectations, and approach to advising fit their goals and ways of working.

- **Research interest alignment.** I look for students whose research interests connect with mine and whose aspirations I am well placed to help them pursue.
- **Strong technical foundations.** My type of research requires a solid foundation in mathematics and computing, together with an interest in using both.
- **A serious commitment.** I take advising seriously because helping a student grow matters deeply to me. I look for students who bring a similar commitment to both the advising relationship and their PhD studies, with a sincere desire to learn and grow and a willingness to invest sustained effort in that process.
- **Character.** I value curiosity, humility, honesty, diligence, reliability, courage, and resilience.

4 What I hope students gradually develop

For me, the primary goal of PhD supervision is to prepare the next generation of educators and researchers in optimization and operations research. The qualities below support that goal and take time to develop.

- **Enjoyment.** I hope students find research exciting and enjoyable. A love of the work itself matters: approaching problems with curiosity and a sense of play can help us do better research.
- **Ambition.** I hope students develop the ambition and confidence to pursue research with lasting value. For me, this means aiming for contributions such as:
  - **Theory:** Fundamental insights, expressed clearly and concisely, that could be taught in PhD-level courses and/or become part of the next generation of textbooks.
  - **Computation:** Methods that are incorporated into widely used optimization solvers, or findings that provide substantive guidance for computational practice.
  - **Applications:** Work that can be implemented in practice and deliver demonstrable benefits for the organizations that use it.

  These are aspirations that guide the choice of research questions. A project can still make a worthwhile contribution even when it falls short of its original ambition.
- **Understanding and clarity.** I hope students develop a fundamental understanding of problems and algorithms. To me, deep understanding also means being able to explain the essential ideas intuitively, with clear logic and concise language.
- **Judgment and independence.** As their experience grows, students can take increasing ownership of their research: propose worthwhile questions, identify promising approaches, break difficult problems into manageable steps, and ultimately advance our understanding in a focused area.
- **Attention to detail.** In my experience, good research requires painstaking, sometimes painful work to clarify details. I hope students develop the patience and discipline to work through tedious or repetitive tasks thoroughly, avoiding superficial contributions, shortcuts, and inflated claims.

5 My responsibilities as an advisor

Helping students develop the qualities described above requires effort and commitment from me as an advisor. The following are some of my current thoughts on the responsibilities this entails.

- **Lead by example.** Most importantly, I should demonstrate in my own work and conduct the qualities (as stated above) that I hope students will develop.
- **Make time for each student.** I should keep my advising commitments manageable so I can give each student meaningful attention. At present, I think a group with at most four or five PhD students and postdoctoral researchers would allow this.
- **Clarify expectations in both directions.** Explain my expectations early and encourage students to express their own interests, goals, and expectations. Help students who are still uncertain begin to articulate what they want.
- **Help students find the right starting point.** For students new to research, offer clear guidance and a well-defined, manageable first problem, and help them become familiar with and get confident in the research process. For students who already have research experience, help identify important, challenging questions that interest us both.
- **Give timely, constructive feedback.** Read students' work and offer specific feedback. Help them understand the reasoning behind my suggestions and identify concrete next steps. Encourage them to question my reasoning and offer their own views.
- **Adapt guidance as students grow.** Students differ in technical preparation, maturity, and clarity of purpose. Adjust the amount and kind of guidance to each person and their stage of development, and give students increasing room to exercise judgment and take ownership of their research.
- **Support students' longer-term development.** Help students identify their strengths, develop their own research directions, and find opportunities and connections that support their goals.
- **Respond with care.** Pay attention to frustration and pressure. Learn to listen and respond constructively when students struggle; emotional support is also part of my responsibility as an advisor.